

CSC8645 / CSC8635 Coursework Report

Task 1: Food Image Segmentation | Task 2: Multimodal Idiomaticity Representation

Task 1: Food Image Segmentation

1.1 Exploratory Data Analysis

The official split files were loaded for FoodSeg103 [1], giving 4,983 training images and 2,135 test images. Each image has a pixel-level mask covering 104 classes — class 0 is background and classes 1–103 are food categories. The training set was divided 80/20, producing 3,986 training images and 997 validation images. Google Colab's high-end GPU configuration was used throughout. A class scan over 300 training masks found 78 of 103 food categories present; background appeared in all 300, confirming class imbalance as the main data challenge.

1.2 Architecture

Both models use U-Net [2] with a ResNet34 encoder via segmentation-models-pytorch. Images and masks are resized to 256×256. The decoder outputs (batch, 104, 256, 256) logits; argmax over the class axis gives per-pixel labels. mIoU is the primary metric — it averages per-class IoU across all 104 classes equally, which keeps it robust to the background imbalance. Pixel accuracy is tracked as secondary.

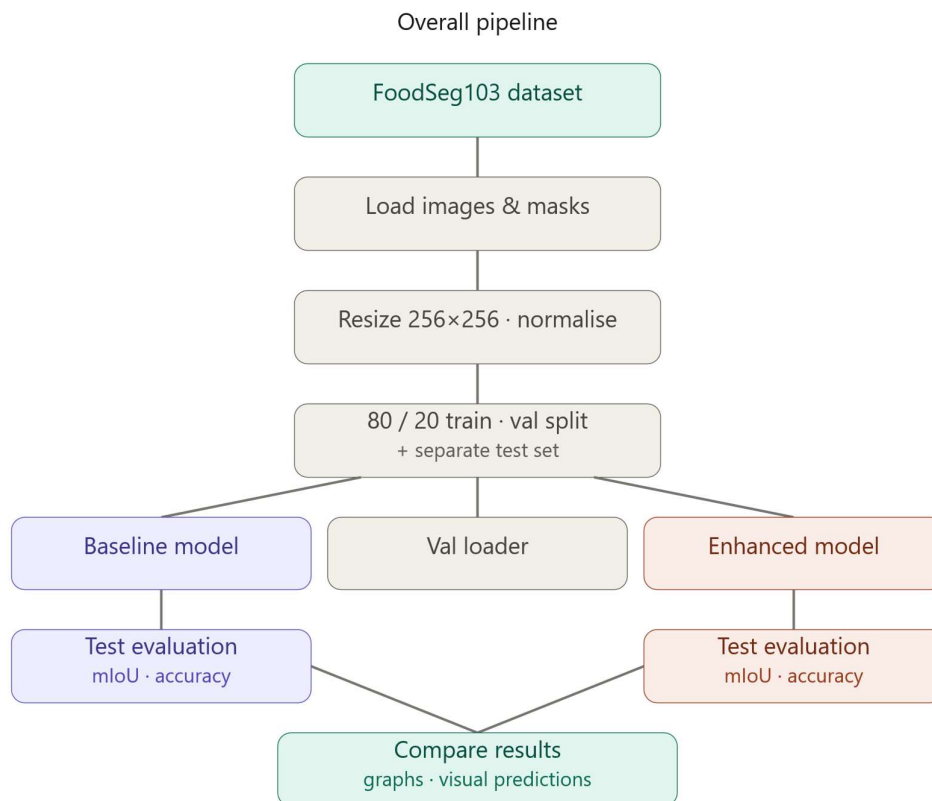


Fig. 1 — Task 1 pipeline: data loading, EDA, model training and evaluation.

1.3 Baseline Model

ResNet34 was initialised randomly with no augmentation and no checkpoint selection. CrossEntropyLoss with no class weights, Adam $\text{lr}=1 \times 10^{-4}$, 10 epochs, batch 32. Val mIoU moved from 0.0219 at epoch 1 to 0.0523 at epoch 10. Test: Accuracy 0.5397, mIoU 0.0523.

1.4 Enhanced Model

The same U-Net + ResNet34 structure was kept, with five additions: (i) ImageNet pre-trained encoder weights [3]; (ii) background class weight set to 0.1, all food classes at 1.0; (iii) ReduceLROnPlateau scheduler (factor=0.5, patience=3); (iv) joint augmentation on image and mask — horizontal flip, 90° rotation, zoom (0.75–0.95), colour jitter; (v) checkpoint saved only on Val mIoU improvement. First save at epoch 1 (Val mIoU 0.0248); 45 epochs in total. Test: Accuracy 0.6829, mIoU 0.2118.

1.5 Results

Model	Accuracy	mIoU
Baseline — random init, 10 epochs	0.5397	0.0523
Enhanced — ImageNet + aug + scheduler, 45 ep	0.6829	0.2118

Pre-training, weighted loss, scheduling, augmentation, and checkpointing together brought about a $\sim 4\times$ mIoU improvement ($0.0523 \rightarrow 0.2118$) and lifted pixel accuracy from 0.5397 to 0.6829.

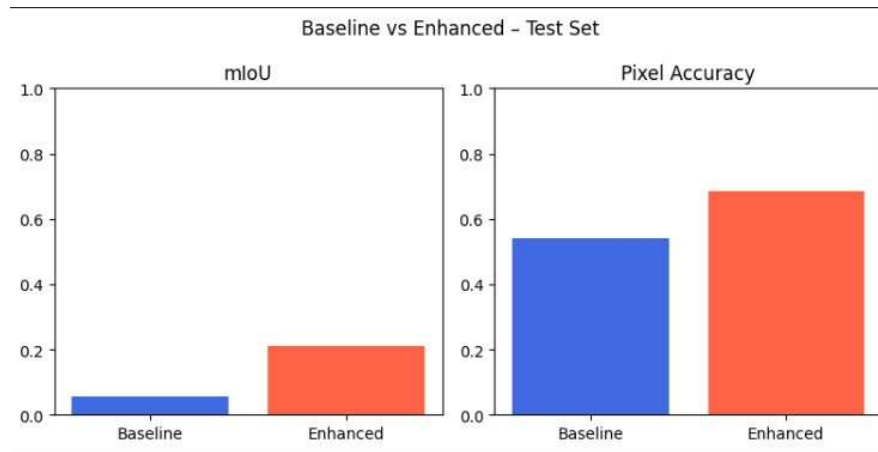


Fig. 2 — Baseline vs. enhanced model on mIoU and pixel accuracy.

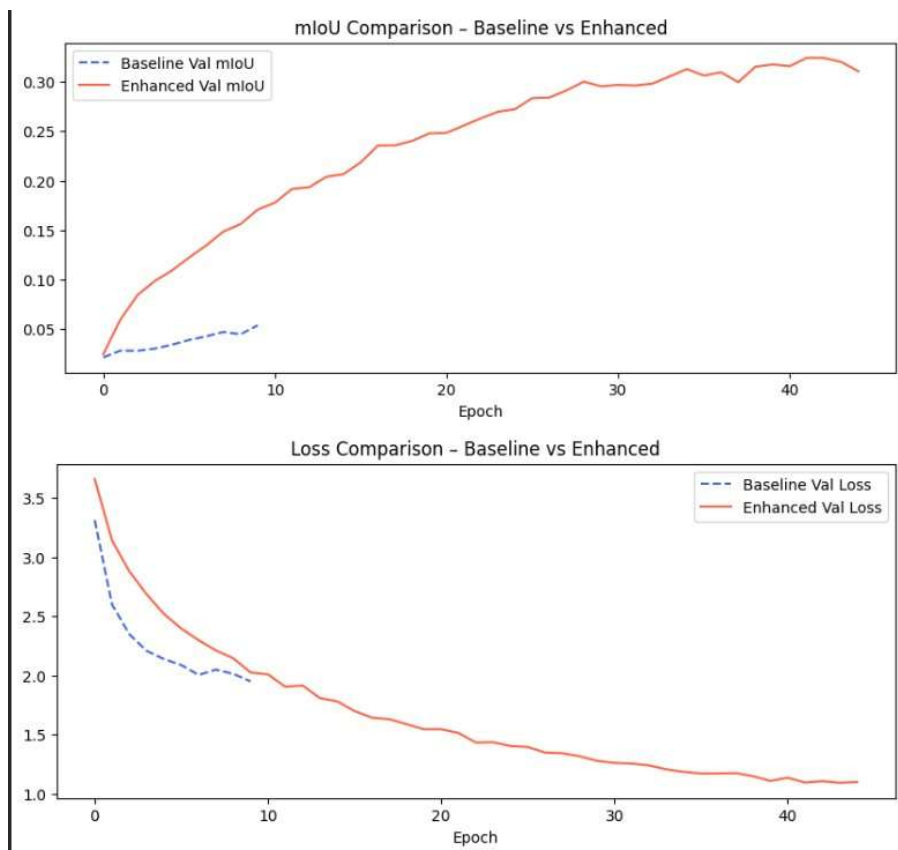


Fig. 3 — Training curves: mIoU and loss comparison across epochs.

References

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- [4] C. Raffel et al., "Exploring the limits of transfer learning with a unified text-to-text transformer," J. Mach. Learn. Res., vol. 21, no. 140, pp. 1–67, 2020.
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